

TrES-1b

R-0232

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_250620 · created 2026-07-17T03:45:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460846.8286 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.034
Transit depth (Rp/Rs)^2	0.92 +/- 0.66 [%]
Orbital Inclination (inc)	85.01 +/- 1.47
Airmass coefficient 1 (a1)	0.852 +/- 0.013
Airmass coefficient 2 (a2)	0.155 +/- 0.062
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#3 - [52, 402]
Optimal Method	PSF photometry
Transit Duration (day)	0.048 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.096 $\pm$ 0.034 vs 0.1358 (deviation 1.17 σ)
Duration fit vs published	0.048 d vs 0.1042 d (deviation 1.7 σ)
Mid-time O–C	7.3 min
Depth SNR	2.8
Residual scatter	1.51 %

Light curve

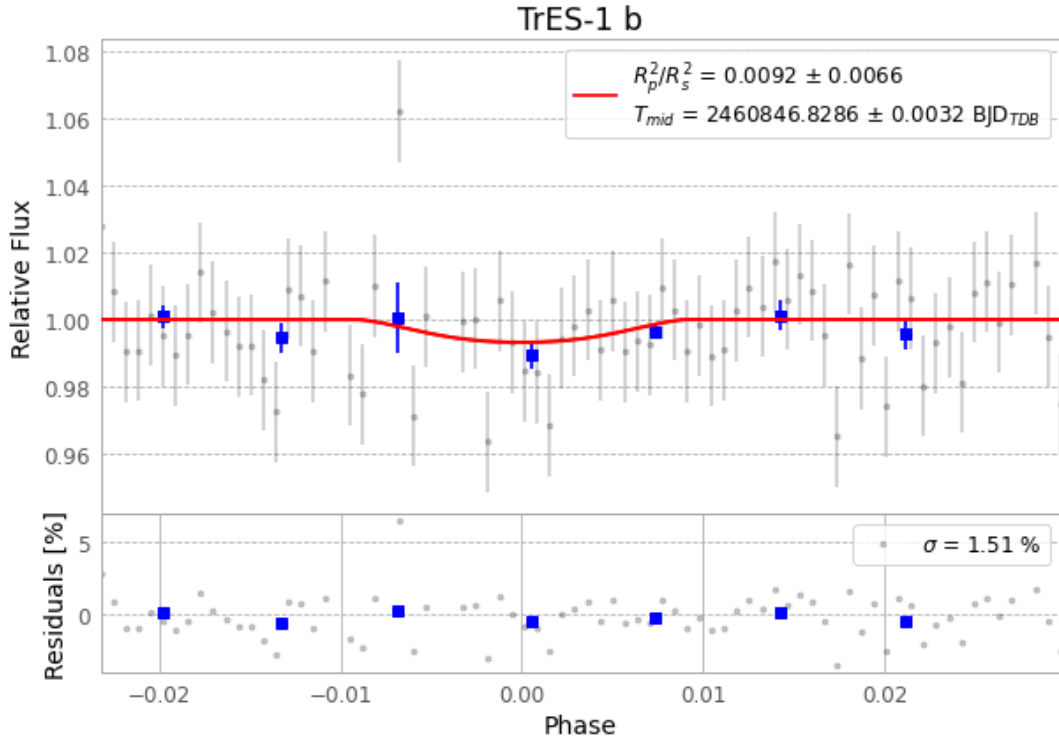


Figure 1 — FinalLightCurve_TrES-1 b_2025-06-19.png (SHA-256 55be9e920ed2abeb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5fb88eda0a6a1cad...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

78 FITS frames (51 MB), TRES-1250620060615.FITS → TRES-1250620095715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250620.log (a079886ef517...), FinalLightCurve_TrES-1 b_2025-06-19.png (55be9e920ed2...), FinalLightCurve_TrES-1 b_2025-06-19.csv (bfbfde0047a4...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 03:45Z.